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Review

Epidemiology and Control of Foodborne Diseases in Resource-Limited Settings: Insights from Afghanistan



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ABSTRACT

Food safety is a major global public health concern, with unsafe foods contaminated by biological, chemical, and physical hazards causing foodborne diseases (FBDs), resulting in a high global burden of illness and death. This narrative review provides an overview of current evidence on the epidemiology, risk factors, and control strategies of FBDs, with particular attention to challenges in resource-limited settings. These diseases are disproportionately severe in low- and middle-income countries (LMICs) due to interconnected factors such as poor sanitation, limited access to clean water, inadequate food safety practices, weak regulatory systems, and socioeconomic constraints, leading to high morbidity, mortality, economic losses, and an especially heavy burden on vulnerable populations like children under five, with countries such as Afghanistan further affected by fragile health systems and widespread informal food markets. This review synthesizes current evidence on the epidemiology, risk factors, and control strategies of FBDs, using Afghanistan as a representative case of a fragile and resource-limited setting. The findings highlight that microbial contamination, particularly from *E. coli*, *Salmonella* spp., and parasitic pathogens, is driven by unsafe water, informal food markets, and poor hygiene practices. Systemic challenges, including fragmented governance, insufficient surveillance systems, and limited infrastructure, further exacerbate the burden. Comparative analysis with other LMICs reveals common structural determinants, suggesting that the challenges observed in Afghanistan are broadly applicable across developing regions. The study emphasizes the importance of integrated approaches, including One Health frameworks, risk-based food safety systems, and targeted behavioral interventions. Strengthening surveillance, improving infrastructure, and enhancing regulatory coordination are critical to reducing the global burden of FBDs. This review provides actionable insights for policymakers, researchers, and international stakeholders aiming to improve food safety in resource-constrained settings.

Food safety is a paramount issue for any nation. Unsafe foods are a source of biological, chemical, and physical contaminants, which threaten human health and the food supply chain. Globally, these diseases signify a substantial public health concern, imposing a significant economic, health, and social burden (Pires et al., 2021). More than 250 different sources of foodborne diseases (FBDs) have been identified, with biological contaminants (such as bacteria, protozoans, viruses, and fungi) being the key offenders that make food unsafe and cause FBDs (Odeyemi, 2016; Zahir, Ge et al., 2025). These pathogens often lead to wide-ranging outbreaks due to the consumption of unsafe foods, which leads to nearly 600 million cases of FBDs and 420,000 deaths annually (Tzachor, 2024). This burden excessively affects children under 5 years old (Onyeaka et al., 2022).

Moreover, more than 200 diseases are transmitted through the consumption of unsafe food and water, which can occur at different phases of the food supply chain, contributing to the FBDs (Zahir, Anis, & Mushfiq, 2025). However, important distinctions exist between foodborne, waterborne, and zoonotic diseases based on their primary transmission pathways. They are primarily transmitted through contaminated foods, whereas waterborne diseases are mainly linked to contaminated drinking water and inadequate sanitation systems (WHO, 2024; CDC, 2025). In contrast, zoonotic diseases are naturally transmitted between animals and humans through direct contact, vectors, or animal-derived products, although some zoonotic pathogens may also be transmitted through food under specific conditions (FAO, 2026a). Clear differentiation among these categories is

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essential for accurate epidemiological interpretation and targeted public health interventions.

The complexity of food safety management is highlighted by such prevalent contamination, which calls for an inclusive, thoughtful consideration of etiologies and transmission pathways (Albert et al., 2024). FBDs have noteworthy financial impacts as well; each year, healthcare costs, lost productivity, and lost commerce amount to billions of dollars (Abdullahi et al., 2025). The annual healthcare expenditures and productivity losses associated with contaminated food in LMICs exceed US$110 billion, underscoring the substantial economic burden imposed by inadequate food safety systems (WHO, 2025).

The substantial impact of FBDs on public health is shown by the fact that their international burden is comparable to that of serious infectious diseases, including malaria, HIV/AIDS, and tuberculosis (Maikanov et al., 2021). The first estimates of the global and regional burden of FBD were released by the World Health Organization (WHO) in 2015. WHO examined 31 foodborne hazards and showed that the impact was similar to that of other significant infectious diseases. Despite these significant global statistics, due to limitations in monitoring systems and diagnostic skills, especially in low- and middle-income countries, the accurate impact is undoubtedly underestimated (Obande, 2023). To support evidence-based policy creation and intervention strategies at the national, regional, and global levels, however, continuous quantification efforts are essential due to the significant impact of FBDs, particularly on vulnerable populations (Keddy et al., 2025).

The complicated interaction of factors such as limited access to potable water, scarce sanitation, and suboptimal food handling practices worsens the prevalence of FBDs in regions, for instance, in Asia and the Global South (Oduoye et al., 2024). Inadequate training on food safety and frail regulatory frameworks exacerbate the situation, making it problematic to determine the real impact (Kaugi et al., 2023). Infectious diseases, including FBDs, continue to be a foremost cause of the worldwide disease burden with high morbidity and mortality in developing countries despite improvements in systems for traceability and inspection measures (Valero et al., 2016).

Implementing and imposing adequate food safety actions are predominantly difficult in these areas due to the widespread complications of food insecurity, poverty, and illiteracy (Subedi et al., 2025). This underscores the perilous need for a comprehensive understanding of the specific drivers and epidemiological patterns of FBDs across diverse settings to effectively inform targeted public health interventions (Al Khatib & Kabir, 2025). In Southeast Asia and Africa, where 125,000 children under the age of five die annually, the global burden of disease caused by foodborne pathogens is especially severe (Salamandane et al., 2023). Compared with people in low- and middle-income countries, this category bears a disproportionate amount of the burden of FBDs globally. They also bear high financial costs, such as medical care costs, effects on child development, lost productivity, and decreased international trade (Keddy et al., 2025). Like many low-income nations, Afghanistan is disproportionately affected by chronic diseases because of factors like poor healthcare systems, lax enforcement of regulations, and widespread unofficial food markets (Abdullahi et al., 2025). Socioeconomic limitations and environmental variables, such as poor sanitation, which worsen the spread of waterborne infections, further strain the already unwarranted healthcare system, which is already burdened by infectious diseases like malaria and tuberculosis. According to national reports, more than 150,000 malaria cases were reported in Afghanistan in 2019. Tuberculosis also remains a major public health concern, with approximately 48,420 cases reported by the National Tuberculosis Program (NTP) in 2019, while an estimated additional 70,000 cases were projected nationwide (Essar et al., 2023).

The possibility of pathogen growth in food products is greatly increased by the environmental susceptibility, which raises the frequency of FBDs (Al Khatib & Kabir, 2025). Widespread poverty, with a sizable section of the population living below the poverty line, and poor literacy rates all contribute to this perilous position by impeding effective public health initiatives and food safety education (Mubarak et al., 2016). These issues fuel a vicious cycle of susceptibility in which tainted food items, often offered in unregulated street markets, become the main carriers of a variety of infections (Zahir, Anis, & Jauhar, 2025). Among the often-reported pathogens in comparable circumstances are *Vibrio cholerae*, *Salmonella*, *Escherichia coli* (*E. coli*), and *Campylobacter*, which increase both immediate medical costs and enduring consequences (Arias-Granada et al., 2021). As a result, these pathogens' persistent prevalence adds significantly to the burden on public health, as seen by acute disease, long-term consequences, and increased medical costs (Chea, 2023). Furthermore, FBDs have significant negative economic effects that go beyond direct medical expenses to include lost productivity, decreased trade, and lower consumer confidence (Severino et al., 2025). Therefore, this study aims to address the current status of FBDs in Afghanistan and compare it with the situation in some developing countries. This study will explore the epidemiology, risk factors contributing to the incidence of FBDs and highlight strategies that play an important role in FBDs' control, providing new information for researchers and policymakers (Fig. 1).

Methodology

Literature search strategy. This study was conducted as a narrative review. It aimed to summarize and synthesize available evidence on the epidemiology, sources, transmission pathways, risk factors, challenges, and control strategies of FBDs in Afghanistan and other resource-limited settings. A structured literature review was conducted to identify and synthesize available evidence related to the epidemiology, sources, transmission pathways, risk factors, challenges, and control strategies of FBDs in Afghanistan and other resource-limited settings. To provide a broader context and facilitate comparative discussion, evidence from LMICs with similar food safety challenges was also considered. Relevant literature was identified through electronic searches of major scientific databases, including Scopus, Web of Science, PubMed, and Google Scholar. Literature published between 2010 and 2026 was considered to capture both earlier and more recent developments in food safety research and public health evidence. In addition, reports and technical documents from international organizations, including the WHO, Food and Agriculture Organization (FAO), and United Nations Children's Fund (UNICEF), were consulted where relevant to obtain publicly available information on surveillance systems, policy initiatives, and food safety-related programs. Relevant literature was identified through electronic searches of major scientific databases, including Scopus, Web of Science, PubMed, and Google Scholar. Literature published between 2010 and 2026 was considered to capture both earlier and more recent developments in food safety research and public health evidence. In addition, reports and technical documents from international organizations, including the WHO, Food and Agriculture Organization (FAO), United Nations Children's Fund (UNICEF), and some other sources, were consulted where relevant to obtain publicly available information on surveillance systems, policy initiatives, and food safety-related programs. The search strategy used combinations of keywords and Boolean operators adapted according to database requirements. Search terms included, but were not limited to, “foodborne diseases”, “food safety”, “foodborne pathogens”, “Afghanistan”, “epidemiology”, “risk factors”, “food contamination”, “street food”, “water contamination”, “antimicrobial resistance”, “One Health”, and “resource-limited settings”. Additional relevant studies were identified through manual screening of reference lists from selected publications.

Eligibility criteria. Eligibility criteria were developed before study selection to improve consistency and ensure that the included literature aligned with the objectives of this review. Studies were considered eligible if they: (i) addressed foodborne diseases, foodborne

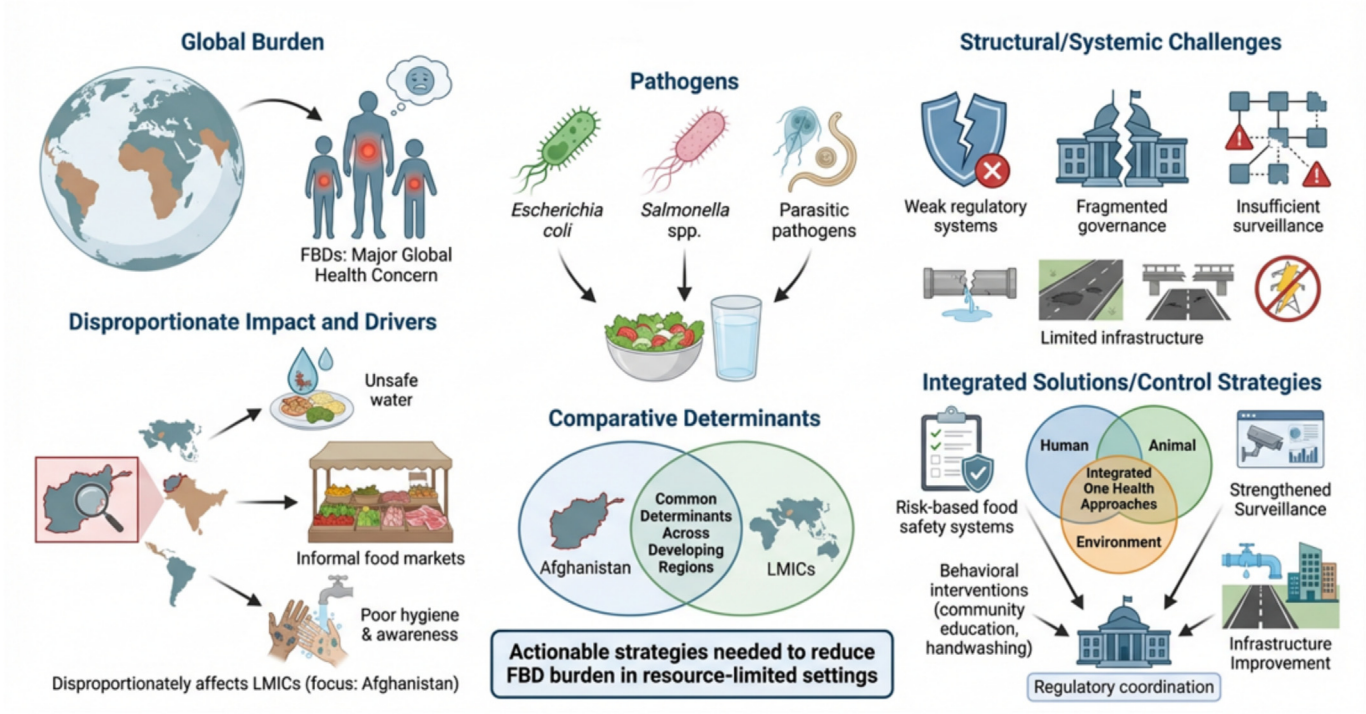


Figure 1. Foodborne diseases (FBDs) remain a major global public health concern, disproportionately affecting low- and middle-income countries (LMICs) due to weak regulatory systems, inadequate sanitation, and limited food safety awareness.

pathogens, food safety practices, or related public health concerns; (ii) focused on Afghanistan or comparable developing and resource-limited settings; (iii) reported findings related to epidemiology, prevalence, transmission pathways, risk factors, surveillance activities, interventions, or food safety policies; (iv) were published as peer-reviewed research articles, review papers, surveillance studies, or reports from recognized international organizations; and (v) were available in English. Studies were excluded if they: (i) consisted solely of conference abstracts without accessible full-text versions; (ii) were not directly relevant to the scope and objectives of this review; (iii) represented duplicate publications; (iv) lacked sufficient methodological or contextual information; or (v) focused exclusively on unrelated laboratory or clinical outcomes without clear relevance to food safety or foodborne diseases.

Study selection and evidence synthesis. Titles and abstracts identified through the database search process were initially screened for relevance. Potentially eligible articles subsequently underwent full-text evaluation to determine their suitability for inclusion. Study selection was guided by relevance to the review objectives, clarity of reported findings, and overall contribution to understanding foodborne disease issues within Afghanistan and comparable settings. Considering the broad scope of the topic and the variability in available evidence, a qualitative narrative synthesis approach was adopted rather than quantitative pooling of findings. The included literature was reviewed and synthesized according to major themes relevant to the objectives of the study. These themes included epidemiological patterns, common pathogens, transmission pathways, risk factors, infrastructure and policy challenges, and existing or potential control strategies.

Quality considerations. The available evidence on FBDs in Afghanistan and comparable resource-limited settings should be interpreted cautiously due to several recurring methodological limitations. Most studies are cross-sectional, limiting causal inference and the assessment of temporal trends. In addition, many studies are geographically concentrated in a small number of provinces or urban centers, reducing the representativeness and generalizability of findings at the national level. Convenience and nonprobability sampling approaches were frequently used, potentially introducing selection bias. Sample sizes also varied considerably across studies, with several investigations relying on relatively small study populations. Furthermore, surveillance and reporting systems remain fragmented and inconsistent, contributing to underreporting and limiting the comparability of epidemiological estimates across regions and over time. Diagnostic methods, laboratory capacity, and outcome definitions also differed between studies, further increasing methodological heterogeneity. Consequently, the evidence base should be regarded as informative but of variable quality, and conclusions should be interpreted as indicative of prevailing patterns rather than precise estimates of disease burden. Future research would benefit from larger multicenter studies, standardized surveillance frameworks, harmonized diagnostic protocols, and longitudinal designs to strengthen the quality and reliability of evidence. Furthermore, the absence of a functioning national foodborne disease surveillance system means that outbreak data are largely passive, underreported, and subject to detection bias. These methodological constraints do not invalidate the synthesized evidence but require that findings be interpreted as identifying recurrent risk patterns and hypotheses for future investigation rather than as precise national burden estimates.

Epidemiology in Afghanistan. The epidemiological profile of Afghanistan is complicated, highlighted by a prolonged burden of infectious diseases, and with fast emerging incidence of noninfectious conditions. The country's health system is sturdily influenced by previous prolonged conflict, immigrants returning from neighboring countries, internal displacement, climate change, and limited access to health care services and infrastructure, all of which have deteriorated illness surveillance and response systems (Zahir & Hamdard, 2025). Infectious diseases, especially malaria, measles, tuberculosis, and acute diarrheal and seasonal respiratory infections, which excessively affect children under 5 years of age, remain the leading cause of morbidity and mortality. According to national reports, Afghanistan

recorded over 150,000 malaria cases and approximately 48,420 tuberculosis cases in 2019, with an estimated additional 70,000 tuberculosis cases nationwide (Essar et al., 2023). Simultaneously, this country continues to face challenges like the endemic poliovirus spread, reflecting gaps in vaccination coverage and healthcare access (Zahir, Anis, & Jauhar, 2025). Heightening this burden is the high prevalence of malnutrition, including stunting, wasting, and micronutrient deficiencies, which not only increases vulnerability to infections but also contributes to prolonged adverse health consequences.

When it comes to infectious diseases, food and water contaminated with feces are the most prevalent way for hepatitis A and hepatitis E to spread. Both illnesses are caused by viruses and affect liver function. Fever, jaundice, and diarrhea are signs of hepatitis A and E, which are common in unsanitary regions. Bacteria like *Salmonella*, *Shigella*, *Campylobacter*, and *E. coli* most frequently cause acute bacterial diarrhea. Three percent of Afghans are thought to be infected with amoebiasis. Up to 11% of the youngsters surveyed had giardiasis. Cholera is another serious diarrheal illness in Afghanistan. The bacteria *V. cholerae* cause an intestinal infection that results in cholera, an acute sickness. Usually asymptomatic and moderate, the infection has the potential to worsen. Vomiting and watery diarrhea are among the severe symptoms that an estimated one in ten cholera patients will experience. Rapid bodily fluid loss can result in shock and dehydration, which could be lethal if left untreated (Common Diseases in Afghanistan, 2026).

In recent years, urbanization, dietary changes, and lifestyle variables like tobacco use and physical inactivity have all contributed to a noticeable increase in noncommunicable diseases, including diabetes, cardiovascular disorders, and some types of cancer. Environmental factors, especially inadequate access to clean water, poor sanitation, and food safety issues, accelerate the spread of foodborne and waterborne diseases. However, because of underreporting of FBDs, inadequate laboratory capacity, and disjointed health information systems, the actual extent of the FBDs burden in Afghanistan is probably underestimated. Despite these obstacles, national and international partners continue working to improve public health infrastructure, expand immunization programs, and bolster epidemiological surveillance. An inclusive understanding of the epidemiological trends in Afghanistan is indispensable for developing targeted interventions, informing policy choices, and addressing the country's evolving public health priorities (Zahir, 2026).

Common pathogens and prevalence data. Globally, foodborne pathogens, including bacteria, viruses, parasites, and fungi, are the main causes of FBDs (Elbehiry et al., 2023). They still cause numerous intestinal diseases in humans, ranging from diarrhea to death (Gourama, 2020). There are 31 major foodborne pathogens that cause FBDs; national authorities monitor the major ones, including *Salmonella* nontyphoidal, *Campylobacter*, *Listeria*, and Shiga toxin-producing *E. coli*. Outbreaks are thoroughly examined to identify trends and determine the actions required to prevent future outbreaks (Adley & Ryan, 2025). The commonly reported FBD outbreaks caused by known pathogens are presented in Figure 2. Among them, two-thirds of FBDs in humans are caused by bacteria, with a high prevalence in developing nations (Abebe et al., 2020). In Afghanistan, several studies have reported that numerous food- and waterborne pathogens contribute significantly to infectious diseases (Table 1). A study conducted in Herat city reported that coliform bacteria were detected in 70% and 75% of beef samples, while *E. coli* contamination rates reached 52% and 72% in samples collected from slaughterhouses and butcheries, respectively (Shakhes et al., 2023). These values suggest relatively high contamination levels, although comparisons with other studies should be made cautiously due to differences in methods and sampling approaches.

An outbreak of acute gastroenteritis occurred in the southern province of Nimroz in late December 2007. Among the 83 probable cases identified, 17 (20.5%) resulted in death, indicating a remarkably high case fatality rate. Epidemiological analysis revealed a significant association between illness and the handling and consumption of camel meat. Molecular investigations using polymerase chain reaction (PCR) combined with electrospray ionization–mass spectrometry detected DNA signatures consistent with *Yersinia pestis* in clinical specimens from patients as well as in camel tissue samples. Despite the atypical gastrointestinal presentation, the diagnosis of plague was confirmed through real-time PCR and serological evidence of immunological seroconversion in at least one patient (Leslie et al., 2011). This represents an unusual food-associated outbreak, highlighting possible zoonotic and atypical transmission pathways in specific contexts.

A study conducted in government and private schools in Jalalabad city, Nangarhar Province, aimed to isolate and identify pathogens from drinking water sources. The findings revealed prevalence rates of *Shigella*, *Salmonella*, and *E. coli* of 63.3, 34, and 30.7% in water tanks, and 60, 38, and 20.7% in wells, respectively (Faisal et al., 2021). Similar contamination patterns have been reported in other developing settings, suggesting persistent water quality and sanitation challenges. In the same city, another study reported that the prevalence of intestinal parasitic infections (IPIs) among schoolchildren was 48.8% (294/603). The most frequently detected parasites included *Giardia lamblia* (35.8%), *Entamoeba histolytica* (34.3%), *Hymenolepis nana* (22.5%), *Ascaris lumbricoides* (5.3%), *Enterobius vermicularis* (1.8%), and *Trichuris trichiura* (0.3%), highlighting a considerable public health burden (Azami et al., 2025).

The results of a cross-sectional study conducted at Wardak Hospital revealed that more than half of the patients diagnosed with giardiasis reported close contact with animals, consumption of unfiltered water, soil exposure, and increased infection rates during warmer seasons (Asad et al., 2024). Similar findings regarding *Giardia intestinalis* infections were also reported from Laghman (7.5%) and Nangarhar (10.6%) provinces (Arabzai et al., 2025). Furthermore, a cross-sectional seroepidemiological study investigating *Toxoplasma gondii* seroprevalence among pregnant women in Kabul demonstrated that 207 out of 431 participants tested positive for immunoglobulin G antibodies against *T. gondii*, corresponding to a seroprevalence of 48.03% (Mosawi et al., 2019). In addition, the prevalence of *G. intestinalis* assemblages A and B in formalin-fixed stool samples collected from 245 Afghan schoolchildren in Ghazni Province was reported to be 21.2% (52 positive samples) (Lass et al., 2017). Variation in prevalence across studies may reflect differences in diagnostic methods, populations, and study settings.

It is important to note that FBDs may also arise from nonmicrobial sources in addition to pathogenic microorganisms. For instance, Kakar et al. (2010) documented an outbreak of Hepatic Veno-Occlusive Disease in western Afghanistan associated with exposure to wheat flour contaminated with pyrrolizidine alkaloids (PAs). Their findings demonstrated that both wheat flour samples and plant extracts collected from the study area contained toxic concentrations of PAs (Kakar et al., 2010). This highlights that foodborne diseases in the region are not limited to microbial agents but also include chemical contamination sources.

Overall, FBDs are mainly caused by microbial pathogens, particularly bacteria, and remain a major global public health concern, with a greater burden in developing countries. Evidence from Afghanistan shows widespread food and water contamination, recurrent outbreaks, and the presence of bacterial, parasitic, and toxin-related agents. These findings highlight persistent food safety risks and the need to strengthen hygiene practices and food control systems.

Comparative perspective: Afghanistan vs other developing countries. As previously highlighted, FBDs continue to impose a substantial burden in LMICs, largely driven by structural weaknesses in governance, infrastructure, and public health systems (WHO, 2015; Adley & Ryan, 2025). However, LMICs are not homogeneous; countries such as Kenya and Zimbabwe demonstrate gradual strengthening of regulatory coordination and food safety governance through incre-

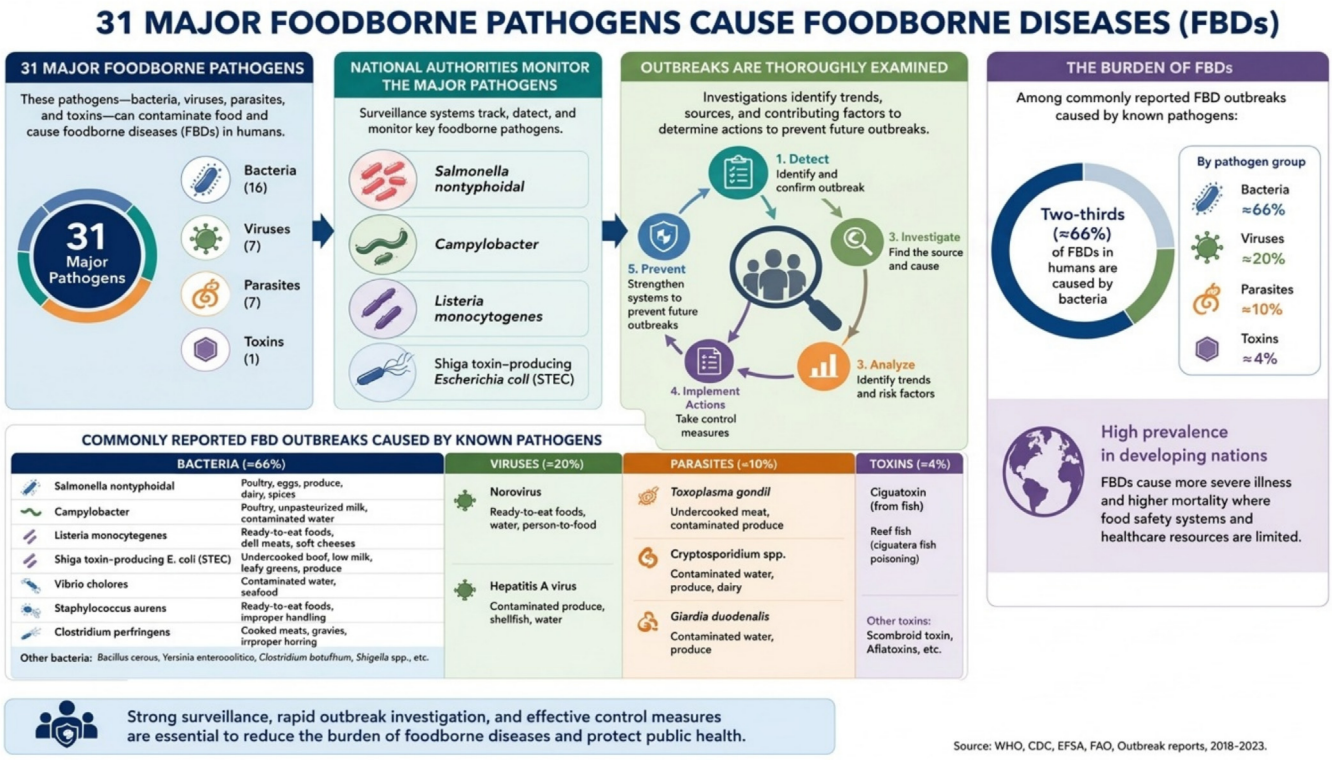


Figure 2. The commonly known pathogens that cause reported FBD outbreaks.

mental institutional reforms, although implementation remains uneven (Kuboka et al., 2024; Mugadza et al., 2025). In comparison, Afghanistan operates under more constrained institutional conditions, where governance fragility limits sustained enforcement of food safety systems and reduces regulatory continuity (Zahir, 2026; Zahir & Hamdard, 2025).

Across LMICs, informal food markets remain a dominant feature of food distribution systems, particularly in Kenya and similar economies where they ensure affordability and food accessibility (Kuboka et al., 2024; Salamandane et al., 2023). In several of these contexts, informal markets are increasingly targeted by partial inspection systems and hygiene improvement initiatives, reflecting gradual institutional integration (Kuboka et al., 2024). In Afghanistan, however, informal markets not only dominate food distribution more extensively but also operate under weaker regulatory reach, with limited inspection frequency and inconsistent enforcement, thereby amplifying contamination risks (Koumassa et al., 2025; Nasiri, 2025).

The knowledge-practice gap in food safety is widely observed across LMICs, including Bangladesh and Nepal, where improvements in awareness have not consistently translated into safe food handling behavior due to socioeconomic constraints and limited institutional reinforcement (Al Banna et al., 2022; Subedi et al., 2025). Evidence from Zimbabwe and other LMICs further indicates that training interventions may improve knowledge but often fail to produce sustained behavioral change without continuous monitoring systems (Mugadza et al., 2025). Afghanistan reflects a more persistent gap, where limited structured training programs and weak supervisory frameworks restrict behavioral compliance among food handlers despite basic awareness of hygiene principles (Zahir et al., 2025; Savant et al., 2025).

Regulatory governance in LMIC food safety systems is frequently characterized by institutional fragmentation, as observed in Zimbabwe and comparable settings where overlapping mandates weaken coordination efficiency (Mugadza et al., 2025). Some LMICs are gradually progressing toward improved regulatory alignment with international standards through institutional reforms (Arero Dassie et al., 2025). In contrast, Afghanistan demonstrates deeper structural fragmentation, where divided institutional responsibilities and weak enforcement mechanisms result in reduced regulatory effectiveness and limited accountability across the food chain (Azizi, 2024).

Infrastructure limitations in cold chain systems, transport logistics, and storage capacity are widely documented across LMICs, including Bangladesh and selected African countries, where gradual improvements are being implemented through targeted investments (Chen et al., 2022; Nabi et al., 2025). Nevertheless, Afghanistan remains significantly constrained by inadequate cold storage capacity, unreliable electricity supply, and underdeveloped logistics networks, which together intensify postharvest losses and microbial contamination risks (Nasiri, 2025; FAO, 2026).

Antimicrobial resistance (AMR) has emerged as a major global food safety threat, closely linked to the misuse and overuse of antibiotics in food-producing animals (Zhao et al., 2024). Weak regulatory enforcement and limited veterinary oversight in LMICs contribute to uncontrolled antibiotic application in livestock production systems, which accelerates the emergence and spread of resistant pathogens (Aminullah et al., 2025; Azizi et al., 2024). A critical constraint is the limited surveillance of antimicrobial use and resistance within food-animal production chains, which reduces the ability to detect emerging resistance patterns and delays effective intervention (Zhao et al., 2024). Evidence across settings also indicates uneven progress in antimicrobial stewardship, where some countries show early regulatory improvements while others continue to rely on fragmented or poorly enforced control systems, highlighting significant governance disparities (Aminullah et al., 2025). Within this context, Afghanistan reflects a higher-risk scenario characterized by weak stewardship implementation, limited veterinary infrastructure, and insufficient

Table 1
Major foodborne pathogens, transmission routes, food vehicles, prevalence, associated risk factors, and public health implications in Afghanistan

Pathogen/hazard	Main transmission routes	Common food vehicles	Reported prevalence/findings in Afghanistan	Associated risk factors	Public health implications	References
Salmonella spp.	Fecal contamination, poor slaughter hygiene, and cross-contamination	Chicken meat, beef, mutton, raw poultry	Fresh chicken contamination reached 79% in Kabul; frozen beef 60% contamination. Ampicillin-resistant *Salmonella* reported in raw chicken meat	Poor transportation hygiene, inadequate refrigeration, lack of HACCP, and weak inspections	Diarrhea, gastroenteritis, antimicrobial resistance (AMR), outbreaks of foodborne illness	(Azimy, 2023; Ahmadi et al., 2024)
E. coli (including *E. coli* O157: H7)	Contaminated water, fecal contamination, and poor dairy hygiene	Raw milk, beef, contaminated water	Presence reported in beef and dairy products; contamination associated with inadequate sanitation and unsafe handling	Unsafe water, poor hand hygiene, and informal slaughtering practices	Severe gastroenteritis, hemolytic uremic syndrome (HUS), and child morbidity	(Zahir and Hamdard, 2025)
L. monocytogenes	Contaminated dairy products, poor cold chain management	Unpasteurized milk and dairy products	Dairy vendors showed limited knowledge regarding infectious disease transmission (<20%)	Absence of refrigeration, poor dairy handling, and lack of awareness	Severe infections among pregnant women, neonates, the elderly, and immunocompromised individuals	(Zahir, 2026)
Campylobacter spp.	Improper poultry handling, contaminated meat	Poultry meat	Likely underreported due to limited surveillance systems	Weak laboratory capacity, inadequate food monitoring systems	Acute gastroenteritis and diarrheal disease	(Khalid, 2016)
Hepatitis A and *E* viruses	Fecal-oral transmission through contaminated food and water	Contaminated water, street-vended foods	Viral *hepatitis* cases remain a major public health concern in Afghanistan	Poor sanitation, unsafe water, overcrowding, and refugee movement	*Hepatitis* outbreaks, liver disease burden	(Essar et al., 2023)
Antimicrobial-resistant bacteria	Misuse of antibiotics in livestock and humans	Meat and animal-derived foods	100% ampicillin resistance was detected among *Salmonella* isolates from chicken meat in Kabul	Unregulated antibiotic use, weak AMR implementation policies	Treatment failure, increased healthcare burden, and resistant infections	(Ahmadi et al., 2024)
Parasitic pathogens (*A. lumbricoides*)	Consumption of contaminated food and water	Raw vegetables, contaminated water	Overall ascariasis prevalence among Kabul University students was 15.8%	Poor sanitation, unsafe water, overcrowded housing	Malnutrition, gastrointestinal complications	(Mubarak et al., 2017)
Acute watery diarrhea (AWD)/ Cholera-related pathogens	Unsafe drinking water, poor sanitation	Contaminated water and food	Over 500,000 AWD cases reported during the 2022 outbreaks	Drought, water scarcity, and poor WASH infrastructure	High morbidity, dehydration, and mortality risk, especially among children	(Essar et al., 2023)
Chemical contamination (iodine deficiency/ heavy metals in salt)	Contaminated food production and processing	Salt and processed foods	Only 3 of 10 salt factories in Kabul met standard conditions	Weak regulation, inadequate quality control	Nutritional deficiencies and chronic toxic exposure	(Pajhwok News, 2025)

monitoring of antibiotic use in production and supply systems, which collectively increase the likelihood of AMR transmission through the food chain (Azizi et al., 2024).

Water, sanitation, and hygiene (WASH) conditions represent another key differentiating factor across LMICs. Countries such as Bangladesh have demonstrated partial improvements in urban WASH systems, contributing to reductions in communicable disease risks in specific settings (Nabi et al., 2025). However, rural and fragile-state contexts continue to experience persistent WASH deficiencies. Afghanistan aligns more closely with the latter category, where limited access to safe water and sanitation services sustains high risks of foodborne and enteric infections (OCHA, 2026; Malik & Akhtar, 2020).

Informal markets remain structurally important across LMICs, including Kenya and other developing economies, where they function as essential food supply channels despite regulatory limitations (Kuboka et al., 2024; Koumassa et al., 2025). In more regulated LMIC environments, incremental improvements in hygiene compliance and inspection coverage are gradually emerging. Afghanistan, however, continues to rely heavily on informal food systems with weaker institutional oversight and limited enforcement capacity, resulting in sustained exposure to microbial and chemical hazards (Nasiri, 2025).

Evidence from LMICs indicates that improvements in food safety outcomes are achievable through coordinated regulatory strengthening, institutional capacity building, and infrastructure development (Mugadza et al., 2025; Arero Dassie et al., 2025). However, effectiveness depends on sustained investment in surveillance systems, logistics, and enforcement capacity across the food supply chain (Chen et al., 2022; Nasiri, 2025). In fragile contexts such as Afghanistan, these systemic requirements are further constrained by governance instability and limited institutional resilience (Zahir & Hamdard, 2025).

Afghanistan shares fundamental structural characteristics with LMICs, including dependence on informal markets, infrastructure limitations, fragmented regulation, and AMR-related risks (Azizi 2024; Grace, 2015). However, in contrast to LMICs undergoing gradual institutional strengthening, Afghanistan demonstrates compounded systemic vulnerabilities where governance fragility and infrastructure deficits interact, resulting in more constrained food safety system performance compared with many peers (Zahir, 2026). The comparative perspective of FBD determinants in Afghanistan vs selected LMICs is presented in Table 2.

Quantitative epidemiological analysis. Foodborne illnesses constitute a major yet often overlooked public health concern in global food systems, with more than 90% of the related health impacts occurring in low- and middle-income countries (Grace, 2023). Reliance on informal food markets in these settings frequently intensifies pathogen transmission, underscoring the need for focused strategies to address widespread food safety vulnerabilities (Rhouma et al., 2024). In Afghanistan, this issue is further exacerbated by insufficient diagnostic monitoring and the rapid growth of complex, unregulated supply chains that facilitate the spread of biological risks (Grace, 2015).

Table 2
Comparative perspective of foodborne disease determinants: Afghanistan vs selected LMIC

Key factor	Afghanistan	Other LMICs, including India, Bangladesh, Nigeria, and Ethiopia	Key insight	References
Water and sanitation (WASH)	Limited access to safe water; reliance on unimproved sanitation; high diarrheal burden	Widespread inadequate WASH infrastructure in rural and peri-urban LMICs	Poor WASH is a major driver of foodborne and diarrheal diseases	(Hu et al., 2019; WHO, 2015)
Street food safety	High dependence on street-vended foods with poor hygiene and weak inspection	The informal street food sector dominates the urban food supply, with frequent contamination.	Street foods are essential but high-risk in LMICs	(Koumassa et al., 2025; Rane, 2011; Ud Din et al., 2022)
Regulatory frameworks	Fragmented governance, weak enforcement, and a lack of risk-based inspection	Weak enforcement and regulatory capacity common across LMICs	Institutional weakness limits effective food safety systems	(Arero Dassie et al., 2025; Mugadza et al., 2025)
Food safety awareness & behavior	Moderate knowledge but poor practice among food handlers and consumers	Similar knowledge–practice gap across LMICs	Behavioral gaps significantly increase contamination risks	(Begum et al., 2025; Savant et al., 2025)
Surveillance systems	Weak, fragmented systems; significant underreporting of FBDs	Limited surveillance and reporting systems common in LMICs	The true burden is underestimated due to weak surveillance	(Grace, 2015; Md Ali & Watanabe, 2025)
Infrastructure (Cold Chain & Labs)	Limited cold storage, weak lab capacity, poor diagnostics	Infrastructure gaps widely reported in LMICs	Poor infrastructure increases contamination risk and delays detection	(Chen et al., 2022; Zimmerman et al., 2018)
Informal food markets	Informal markets dominate with minimal oversight	Informal markets widespread and poorly regulated in LMICs	The informal sector is essential but high-risk for food safety	(Koumassa et al., 2025; Kuboka et al., 2024)
Antimicrobial resistance (AMR)	Misuse of antibiotics in livestock; weak regulation	Rising AMR linked to food systems across LMICs	AMR is a major global food safety threat	(Azizi et al., 2024; Franklin et al., 2022; Zhao et al., 2024)

Although comprehensive assessments of national health outcomes are scarce, data from neighboring areas indicate that enteric agents such as *Campylobacter* spp. and *Salmonella enterica* play a substantial role in morbidity linked to tainted informal food networks (Grace et al., 2024; Havelaar et al., 2022). Moreover, the absence of strong food safety systems in conflict zones like Afghanistan reflects wider regional trends, where poor sanitation and environmental pressures markedly elevate rates of diarrheal infections and their corresponding Disability-Adjusted Life Years (DALYs) (Kirk et al., 2015). In particular, the widespread occurrence of *E. coli* and *V. cholerae* highlights a regional health emergency in which seasonal climatic variations and temperature-driven changes heighten transmission risks for these enduring enteric pathogens (Almulla et al., 2025).

In addition to these biological hazards, indirect expenses such as reduced productivity and enduring effects of chronic conditions significantly magnify the economic toll, which remains largely unmeasured owing to major deficiencies in national laboratory capabilities and disjointed regulatory oversight (Abdullahi et al., 2025). This economic strain, alongside annual global productivity losses surpassing US $110 billion in low- and middle-income areas, calls for a transition to evidence-driven risk evaluation to better address at-risk groups (Chea et al., 2022). Given the marked shortage of high-caliber surveillance information in such environments, an immediate need exists to implement a “One Health” strategy that unifies human clinical health, environmental hygiene, and agricultural production routes (Alarcon et al., 2021). This integrated model is vital for tackling the systemic weaknesses in informal livestock supply chains, where inappropriate antimicrobial use and deficient agricultural methods commonly lead to contamination (Seyoum et al., 2024). The food safety governance challenges across LMICs are depicted in Figure 3.

According to the most recent data available, FBDs remain a significant but poorly measured public health burden in Afghanistan. Through nationwide outbreak surveillance, more than 130,890 cases of acute watery diarrhea and 62 related deaths were reported in 2024, underscoring the ongoing impact of enteric infections transmitted through food and water (AWD Outbreak in Afghanistan, 2026). Despite a significant decline in overall infectious disease DALYs since 1999, enteric and other infectious diseases continue to contribute substantially to morbidity, mortality, and disability, according to recent analyses based on the Global Burden of Disease (GBD) database, even though Afghanistan lacks specific national estimates for FBDs. Although accurate national estimates of the economic cost of FBDs are currently unavailable, the costs are expected to be high due to healthcare costs, lost productivity, lower labor force participation, and restrictions on food exports and commerce. Therefore, to promote evidence-based food safety regulations and to measure the incidence, mortality, DALYs, and economic impact of foodborne infections in Afghanistan, thorough studies are still desperately needed (Khan & Tidman, 2026).

In conclusion, FBDs remain a critical yet underrecognized public health challenge in Afghanistan and other LMICs, driven by informal food systems, weak surveillance, poor sanitation, and inadequate regulatory oversight. The growing burden of enteric pathogens, combined with substantial but largely unquantified economic losses, highlights the urgent need for evidence-based food safety interventions and strengthened laboratory capacity. Implementing an integrated “One Health” approach is essential to improve surveillance, reduce contamination risks, and enhance food safety across human, environmental, and agricultural sectors.

Sources and transmission. FBDs originate from various sources and spread through multiple pathways in the food supply chain, making them a leading global public health concern. Animal-based foods such as meat, poultry, eggs, and dairy are the main sources globally. These foods are often contaminated with pathogens like *Salmonella* spp., *Campylobacter* spp., and *Listeria monocytogenes*. Moreover, fresh fruits and vegetables are a noteworthy source of infection. They are frequently contaminated by soil, irrigation water, or postharvest handling and are often linked with pathogens like norovirus and *E. coli*. The spread of foodborne bacteria is also facilitated by contaminated water used in agriculture and food processing (Rodrigues et al., 2020). Food handlers further contribute to contamination, especially when they practice poor personal hygiene, enabling viruses like hepatitis A and norovirus to spread. Besides cross-contamination between raw and ready-to-eat foods, improper cooking, inadequate storage, and person-to-person transmission via the fecal–oral route are primary pathways for foodborne pathogen transmission. The common transmission pathways of foodborne pathogens are presented in Figure 4. The WHO estimates that contaminated food causes 420,000 deaths

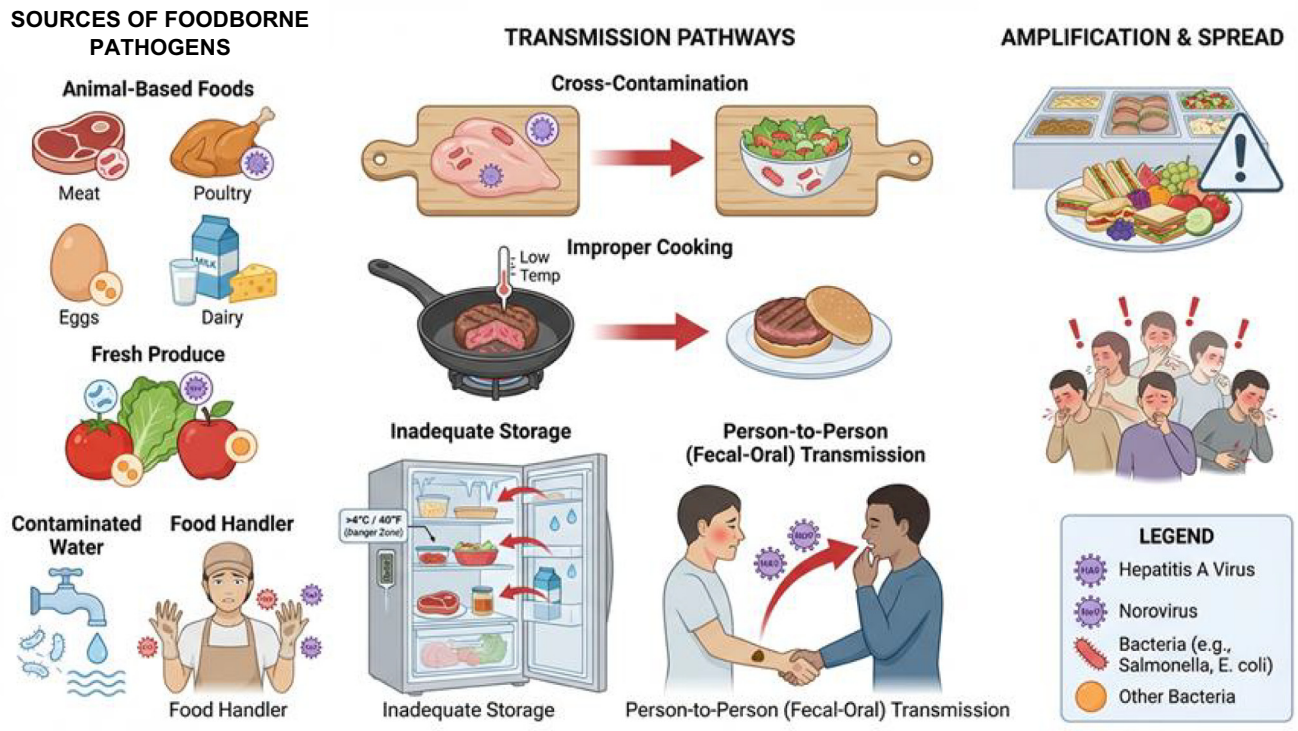


Figure 3. Food safety governance challenges across low- and middle-income countries (LMICs).

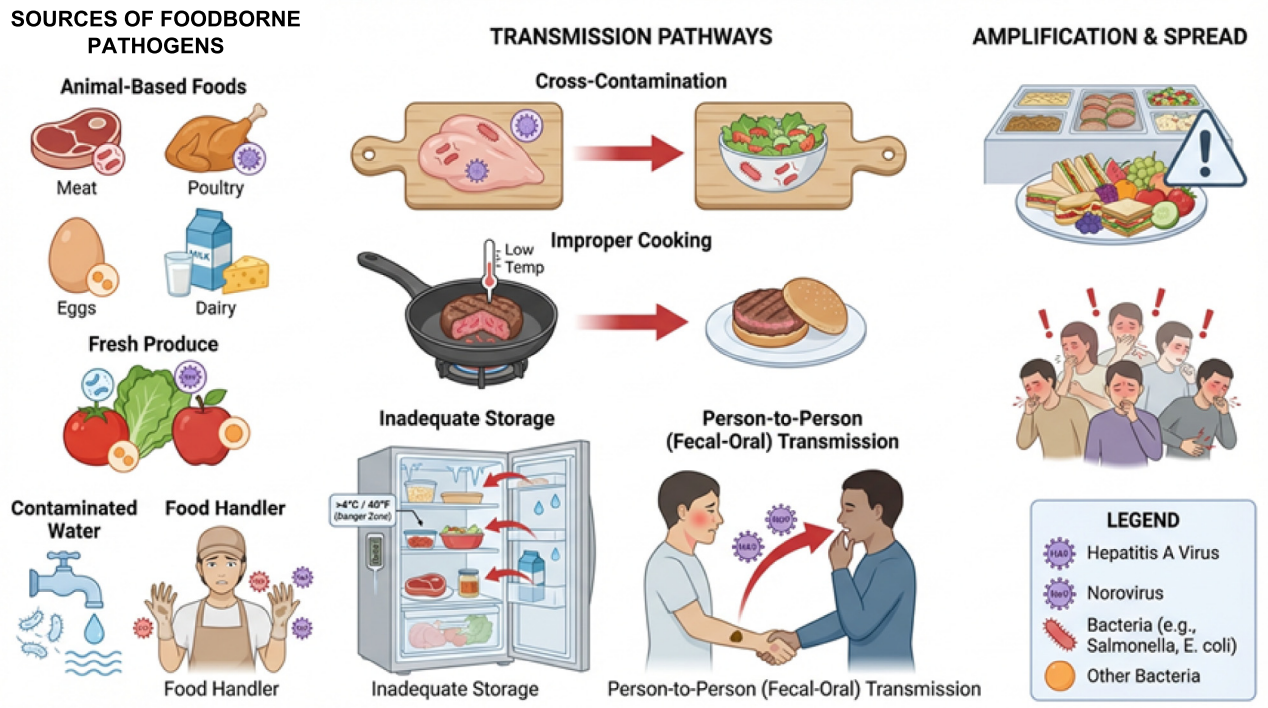


Figure 4. The common transmission pathways of foodborne pathogens.

and 600 million cases of FBDs annually, highlighting the serious burden of these diseases and the need for comprehensive food safety efforts (Zahir, Anis, & Jauhar, 2025).

In Afghanistan, the sources and transmission routes of FBDs are identical to those universal but are worsened by regional issues such as poor sanitation, lacking infrastructure, and limited public knowledge about food safety procedures (Zahir, 2026). Foods from animals remain a major contamination source, especially due to inadequate veterinary oversight, unofficial slaughtering practices, and weak cold chain systems. Studies from Afghanistan indicate substantial contamination of animal-derived foods, particularly meat products, with foodborne pathogens. In Herat Province, coliform bacteria were detected in

70% and 75% of beef samples collected from slaughterhouses and butcheries, respectively, while *E. coli* contamination reached 52% and 72% in the same settings. Additional studies identified pathogens such as *Salmonella* spp., *Shigella* spp., *Staphylococcus aureus*, and coliform bacteria in meat and poultry products, including contaminated chicken livers in Nangarhar Province linked to diarrheal diseases. These findings highlight significant microbiological contamination risks in Afghanistan's animal food supply chain, likely associated with inadequate hygiene practices, weak quality control systems, and limited implementation of food safety measures such as Hazard Analysis and Critical Control Points (HACCP) (Zahir, Anis, & Mushfiq, 2025).

The use of untreated wastewater for irrigation and unsanitary handling in local markets often contaminates fresh produce. In Afghanistan, where access to clean drinking water and proper sanitation remains limited in many areas, waterborne transmission is particularly significant, increasing the risk of fecal, oral pathogen spread (Zahir & Anis, 2026). Moreover, food handlers in both household and street food settings often lack formal training in hygiene practices, heightening the risk of cross-contamination and outbreaks (Zahir, Irfan, & Anis, 2025). Environmental pollution, including dirty surfaces used for food prep and poor waste management, further supports pathogen persistence and spread (Zahir et al., 2024). As a result, FBDs remain underreported but are recognized as a serious public health issue, underscoring the urgent need for better monitoring systems, food safety education, and risk-based food management technique uses. Here, we elaborate on the common source and transmission pathways of foodborne pathogens.

Water contamination. Water contaminated with foodborne pathogens and used for washing of food items and for irrigation of fruit and vegetable crops is the key source and transmission route of foodborne pathogens (Rodrigues et al., 2020). Water-borne diseases, such as diarrhea, are predominantly linked with water contaminated with pathogens, as this issue is reflected in a study conducted by Agha et al. (2024). A study reported 29% diarrhea prevalence in children below 5 years in Afghanistan (Malik & Akhtar, 2020), linked with contaminated water, inadequate sanitation, and shared toilet facilities. However, reported prevalence varies across studies, reflecting differences in study design, population coverage, and water quality assessment methods, which limit the direct comparability of epidemiological estimates. Similar variability in diarrheal disease burden has also been reported across LMICs, where differences in surveillance systems and WASH infrastructure influence observed prevalence patterns (Nabi et al., 2025).

Cholera, caused by *V. cholerae*, is primarily classified as a waterborne disease associated with the consumption of fecally contaminated water and inadequate sanitation infrastructure (WHO, 2024a). Although contaminated foods may contribute to outbreaks under poor hygienic conditions, water remains the principal transmission vehicle in endemic settings such as Afghanistan (CDC, 2025a). Similarly, several intestinal parasitic infections, including *G. intestinalis* and soil-transmitted helminths, are predominantly associated with contaminated water, inadequate sanitation, and poor hygiene practices rather than classical foodborne transmission alone (WHO, 2023). Therefore, these infections should be interpreted within a broader fecal-oral transmission framework that includes both waterborne and food-associated pathways.

A survey conducted in 2017 among schoolchildren aged 8–10 years reported that 26.6% had at least one soil-transmitted helminth (STH) infection, showing a decline compared to earlier reports. *A. lumbricoides* was the most common (25.7%), followed by hookworms (0.1%) and *T. trichiura* (1.0%). All positive cases were light-intensity infections, and coinfection was rare (0.2%) (Safi et al., 2019). In contrast, a study in Ghazni and Parwan provinces reported higher infection burdens, with *G. intestinalis* and *H. nana* as dominant parasites (Korzeniewski et al., 2017). These contrasting findings suggest a declining trend in infection intensity in some settings, although persistent regional heterogeneity indicates uneven improvements in water and sanitation conditions. Comparable heterogeneity is also observed in other LMIC settings, where reductions in helminth burden are more evident in areas with improved sanitation coverage, while rural and informal settings continue to experience high transmission (Arero Dassie et al., 2025). Similarly, moderate-to-high burdens of giardiasis have been reported in clinical settings, where risk factors included unfiltered water consumption and poor hygiene practices (Asad et al., 2024). However, most available studies are cross-sectional and hospital-based, limiting generalizability and weakening causal inference between water contamination and infection burden. In contrast, LMIC-wide evidence indicates that strengthening of WASH systems and improved monitoring in some countries has contributed to reduced infection severity, although progress remains uneven across regions (Nabi et al., 2025).

Taken together, contaminated water remains a major transmission route for enteric and parasitic infections in Afghanistan, particularly among children. Despite some reported reductions in infection intensity over time, intestinal parasites remain widely distributed, indicating persistent gaps in water safety, sanitation infrastructure, and hygiene practices. This pattern reflects a more persistent transmission environment compared with several LMIC settings where gradual improvements in WASH infrastructure have contributed to measurable reductions in water-related disease burden (Mugadza et al., 2025; WHO, 2015).

Street foods. Ready-to-eat street-vended foods are the potential sources for the occurrence of FBDs. A study found unsatisfactory levels of aerobic bacterial counts (>105 CFU/g; 4.3×105 CFU/g), including *Enterobacteriaceae*, coliform, and staphylococcal, and *Salmonella* species in commonly consumed street-vended foods (Alelign et al., 2023). However, reported contamination levels vary across studies, likely due to differences in sampling design, microbiological methods, and hygiene assessment criteria, which limit direct comparability and interpretation of burden. Similar microbiological contamination patterns have been reported across LMICs, particularly within informal food markets where inadequate hygiene infrastructure and inconsistent food handling practices contribute to elevated contamination risks (Kuboka et al., 2024; Koumassa et al., 2025). In Kenya, informal public markets have been identified as important food access points but remain constrained by food safety challenges associated with limited oversight and infrastructure gaps (Kuboka et al., 2024).

Pathogen contamination has also been documented beyond street vending in other informal food environments. For example, *E. coli* contamination was detected in eggshells (43.3%) and egg contents (18.0%) in Kabul (Jaheed & Ahmadi, 2026), while *Shigella* and *Salmonella* were identified in fast foods such as burgers and fried chicken (Weqar et al., 2020). In comparative terms, these findings indicate that contamination levels are not isolated to street foods but appear consistently across multiple informal and semi-formal food outlets, although prevalence estimates differ by food type and setting. Comparable contamination profiles have also been reported in other LMIC contexts, where foodborne pathogens are detected across traditional markets and small-scale retail systems rather than being confined to a single food source (Arero Dassie et al., 2025). Afghanistan, therefore, appears to reflect a broader LMIC pattern, although weaker institutional monitoring may increase the likelihood of under-detection and delayed intervention (Zahir & Hamdard, 2025).

This pattern suggests a persistent contamination risk across informal food markets rather than a single-source problem; however, the absence of standardized national surveillance and the predominance of cross-sectional microbiological studies limit the ability to assess temporal trends or infer true epidemiological progression. These limitations represent a key gap in evidence quality and food safety monitoring systems. Similar surveillance limitations have been documented across several LMICs, where fragmented monitoring systems and inconsistent reporting frameworks reduce cross-country comparability

and hinder evidence-based food safety interventions (Mugadza et al., 2025).

Animal products. Foodborne pathogens are frequently transmitted through animal-derived foods, including milk, dairy products, meat, fish, eggs, and honey (Andrade et al., 2025). Contamination may occur through environmental sources and handling practices across slaughtering, processing, and storage stages (Heredia & García, 2018). Across studies, contamination levels are generally higher in informal markets compared to regulated or semi-regulated supply chains, although the magnitude varies depending on hygiene control and inspection intensity. However, variability in hygiene practices and regulatory enforcement across production systems contributes to inconsistent contamination levels across studies and regions. Comparable patterns have been reported across LMICs, where informal food markets frequently exhibit greater contamination risks due to weaker infrastructure and inconsistent regulatory oversight (Kuboka et al., 2024).

High pathogen prevalence has been reported in informal markets, including *Salmonella* spp. (62.5%), *L. monocytogenes* (27.5%) and diarrheagenic *E. coli* (12.5%) in fishery and meat products, with universal contamination observed in fish fillets (Ribeiro-Júnior et al., 2026). In Afghanistan, raw milk contamination has been reported at 42.33% (Ahmadi, 2025), while *S. aureus* with antibiotic resistance patterns has been detected in dairy products (Ahmadi et al., 2021). These findings indicate significant microbiological risk in both domestic and informal market supply chains, although variations in laboratory methods may affect prevalence estimates. Notably, differences in contamination estimates across studies highlight methodological variability rather than consistent pathogen distribution patterns. Similar variability has also been observed across LMIC settings, where differences in inspection systems and food safety governance complicate direct comparison of contamination burdens between countries (Arero Dassie et al., 2025).

For meat products, *Salmonella* prevalence varies widely across studies, ranging from 8% in chicken meat (Ahmadi et al., 2024) to higher contamination levels in poultry liver and fast market samples (Mohammad et al., 2020). Such discrepancies indicate conflicting findings across similar food categories, likely reflecting differences in sampling frames, processing hygiene, and market conditions rather than true epidemiological stability.

Beyond food consumption, direct animal contact also contributes to zoonotic transmission. For instance, Crimean-Congo Hemorrhagic Fever (CCHF) is primarily a tick-borne zoonotic disease rather than a classical FBD (CDC, 2025b). Human infections mainly occur through tick bites or direct contact with blood, tissues, or bodily fluids of infected livestock during slaughtering or animal handling (WHO, 2025b). Although occupational exposure among butchers and livestock workers may overlap with food production environments, CCHF transmission is epidemiologically distinct from conventional foodborne transmission pathways (Qaderi et al., 2021). Therefore, CCHF should be interpreted within the broader One Health and zoonotic disease context rather than as a typical foodborne pathogen (FAO, 2026a). Similarly, high seroprevalence of *Brucella* and *Coxiella burnetii* has been reported in livestock-owning households (Akbarian et al., 2015). Self-slaughter practices during religious festivals have also been identified as a key risk factor for CCHF transmission (Hamdard et al., 2025). However, most zoonotic transmission studies rely on outbreak investigations or seroprevalence surveys, limiting the ability to quantify long-term population-level risk.

Despite relatively high reported food safety awareness among meat and dairy handlers, significant gaps remain in specific knowledge of FBDs (Zahir, Anis, & Mushfiq, 2025; Zahir, Irfan, & Anis, 2025; Zahir & Anis, 2026). This discrepancy between general awareness and applied food safety behavior represents a persistent behavioral gap in food hygiene practices. However, the absence of standardized surveillance systems and longitudinal datasets limits robust interpretation of exposure trends and long-term risk evolution.

Risk factors

Poor hygiene. In low- and middle-income countries, maintaining the quality and safety of street-vended foods (SVFs) remains a major challenge and a public health priority due to their widespread availability, affordability, and association with microbiological hazards and FBDs (Al Banna et al., 2022). In Afghanistan, butcheries, dairies, and other food retail outlets represent critical points of concern within the food safety system. A study conducted on 182 butcheries in Herat city revealed that 39.0% of butchers were aged 26–35 years, 40.1% were illiterate, and only 3.8% had attained higher education, indicating that individuals directly handling meat often lack formal education in hygiene practices. Although 83.0% of butchers possessed a butchering license, 72.0% had not received any hygiene training, highlighting a significant gap between regulatory compliance and practical capacity. Evidence suggests that practical training is more effective than theoretical approaches in influencing behavioral change (Mohammadi et al., 2024). Similarly, a cross-sectional study of dairy wholesalers and retailers in Kandahar city reported that 92% of vendors demonstrated good knowledge of food safety; however, they lacked awareness of infectious diseases, suggesting that only knowledge is insufficient to drive safe practices (Zahir, Anis, & Mushfiq, 2025).

Microbiological assessments further underscore these concerns. In Kabul Province, *Salmonella* contamination in beef, mutton, and chicken samples exceeded permissible limits, with the highest levels observed in chicken wings (District 5), fresh chicken (District 17), and frozen beef (District 13). These findings were largely attributed to poor hygienic conditions during transportation from slaughterhouses and distribution centers (Ferdows, 2023).

Environmental factors further compound food safety risks. Limited access to potable water and inadequate sanitation infrastructure remain pressing challenges, particularly in rural areas. Contaminated water and poor sanitation are major contributors to diarrheal mortality (Malik & Akhtar, 2020). A multisectoral needs assessment reported that only 15% of the population in western Afghanistan (Herat, Ghor, Badghis, and Faryab) has access to improved sanitation facilities, while 83% rely on basic pit latrines that allow environmental contamination (World Vision, 2025).

Parasitic infections also reflect the broader public health implications of poor hygiene and sanitation. A study among Kabul University students reported an ascariasis prevalence of 15.8%, with recent urban migrants living in shared accommodations identified as high-risk groups. Recommended interventions include promoting safe food preparation practices and ensuring the use of potable water (Mubarak et al., 2017). Overall, these findings highlight systemic challenges in Afghanistan's food safety landscape, including inadequate training, weak enforcement of standards, infrastructural limitations, and a persistent gap between knowledge and practice.

Lack of awareness. Lack of public awareness is a major contributing factor to the spread of FBDs in Afghanistan. Food consumers in Kandahar city demonstrated a moderate level of food safety knowledge (mean score: 10.44 out of 18), with younger consumers (18–25 years) showing less awareness than older age groups (36–45 years). Notably, 76% of respondents expressed uncertainty about food safety when purchasing SVFs, primarily due to convenience (34.4%). Although 95.80% of meat handlers reportedly possess a high level of food safety knowledge, 65.3% do not fully comprehend food poisoning (Zahir, Anis, & Jauhar, 2025). Similarly, less than 20% of dairy vendors in Kandahar were aware of the transmission pathways of infectious diseases, which is particularly concerning given that improperly handled dairy products can provide favorable conditions

for pathogens such as *Salmonella*, *Listeria*, and *E. coli* O157:H7 (Zahir, Anis, & Mushfiq, 2025).

According to a study on Afghan consumers' perceptions of meat quality and safety, although consumers generally express concern about meat safety, they lack sufficient knowledge regarding proper handling and storage practices (Nasiri et al., 2023). To address these gaps, researchers recommend the implementation of targeted training programs aimed at improving both environmental and personal hygiene. Such programs should be practical rather than purely theoretical, culturally appropriate, delivered in local languages, and accessible to individuals with limited literacy. Public health campaigns and media engagement are also critical components of food safety education in Afghanistan, particularly in addressing broader challenges related to food security, nutrition, and public health (Zahir, 2026). Overall, improving food safety and nutrition in Afghanistan requires a comprehensive approach that strengthens regulatory frameworks, promotes public awareness, and strategically utilizes media to enhance public health outcomes, improve food security, and reduce the burden of FBDs and malnutrition.

Weak regulation. A significant obstacle to food safety in Afghanistan is the weak enforcement of laws and regulations. Despite the existence of food safety regulations, their implementation is largely limited to major cities such as Kabul, Herat, Jalalabad, Balkh, Khost, and Kandahar. Three main government bodies are responsible for monitoring food safety in Afghanistan, including the Ministry of Public Health (MoPH), the Ministry of Agriculture, Irrigation, and Livestock (MAIL), the Afghanistan National Standards Authority (ANSA), and provincial municipalities (Zahir, 2026). However, these institutions often lack coordination, leading to duplication of efforts and increased administrative burdens for food business operators. This lack of cooperation remains a major barrier to effective food safety enforcement (Khalid, 2016). In line with international food safety standards, inspections of food establishments should be risk-based; however, Afghanistan has not yet adopted this approach. Instead, all establishments are inspected uniformly, regardless of their level of operational risk.

Another major weakness in the country's food safety system is the limited capacity of testing and laboratory services. Food sampling programs are inconsistent, and oversight of imported food products remains inadequate. Furthermore, Afghanistan has not implemented the globally recognized food safety system, such as HACCP, leaving producers and vendors without structured mechanisms to ensure food safety (Khalid, 2016). Similarly, although 83% of butcheries are licensed, there is insufficient monitoring to ensure compliance with hygiene standards (Mohammadi et al., 2024). The National Action Plan (NAP) on AMR is also poorly implemented. Antibiotics remain widely available without prescription, and their use in livestock production to promote growth contributes to the emergence of antibiotic-resistant bacteria, which can be transmitted to humans through food (Zahir & Hamdard, 2025). Moreover, approximately 8 million Afghan refugees living in neighboring countries such as Pakistan and Iran have been reported to have high rates of hepatitis B (HBV) and hepatitis C (HCV) infections. Upon returning to Afghanistan, these individuals may unknowingly contribute to the transmission of these infections, further exacerbating the country's public health challenges (Essar et al., 2023).

Unlike hepatitis A and hepatitis E viruses, which are commonly transmitted through fecally contaminated food and water, HBV and HCV are primarily bloodborne pathogens transmitted through unsafe injections, contaminated blood products, and other parenteral exposures (WHO, 2025; WHO, 2025a). Therefore, HBV and HCV should not be classified as classical foodborne diseases, although they remain important public health concerns in Afghanistan. Food safety in Afghanistan is significantly hindered by weak enforcement of regulations, limited institutional coordination, and inadequate inspection systems. Despite existing laws, their implementation is inconsistent, and the absence of risk-based inspections and standardized systems like HACCP reduces effectiveness. Additional challenges include limited laboratory capacity, poor monitoring of food businesses and imports, misuse of antibiotics contributing to antimicrobial resistance, and broader public health risks linked to population movement, altogether highlighting serious gaps in the country's food safety framework.

Current challenges

Infrastructure. The lack of cold storage facilities represents one of the most significant infrastructural barriers to food safety in Afghanistan. Although the country produces large quantities of fruits, vegetables, and dairy products annually, these commodities are highly perishable and often spoil due to inadequate cold storage systems. Insufficient refrigeration creates favorable conditions for the rapid growth of bacteria such as *Salmonella* and *E. coli* (Nasiri, 2025). In response, the MAIL has announced the construction of eight new cold storage facilities, each with a standard capacity of 5,000 tons, across six provinces. Additionally, China plans to establish three more facilities, each with a capacity of 1,500 metric tons, in Kabul and Balkh provinces in collaboration with the MAIL (DID Press, 2026). Despite these developments, current processing capacity meets less than 1% of national demand. Consequently, most farmers and livestock producers are compelled to sell their products without refrigeration, significantly increasing the risk of contamination and leading to substantial postharvest losses (Nasiri, 2025).

Laboratory and testing capacity are also highly limited and concentrated primarily in major urban centers. In many cases, food samples can only be analyzed in Kabul, resulting in insufficient surveillance of FBDs in rural and remote areas. This limitation contributes to underreporting and delayed detection of contamination events (Khalid, 2016). Furthermore, access to clean water and reliable electricity remains inadequate in many parts of Afghanistan. The lack of water restricts essential hygiene practices such as handwashing, cleaning of equipment, and safe food preparation, while unreliable electricity supply hinders the operation of refrigeration systems and the maintenance of appropriate storage temperatures (World Vision, 2025).

The transportation infrastructure presents additional challenges. Poor road conditions, high ambient temperatures, and the absence of refrigerated transport systems increase the likelihood of food contamination during long-distance distribution, particularly in rural areas (Nasiri, 2025). Overall, these infrastructural deficiencies are compounded by regulatory gaps, including outdated legislation, weak surveillance systems, and the absence of risk-based food safety management approaches. Moreover, a large proportion of informal market actors operate outside regulatory oversight, further elevating the risk of FBDs across the Afghan food supply chain.

Policy gaps. In Afghanistan, the development of food and nutrition security policies has been fragmented and prescriptive rather than needs-based, resulting in two overlapping policy documents developed independently by different entities. These documents lack consensus, clear ownership, and political commitment for implementation, leading to strategic duplication and ambiguity in decision-making. Consequently, despite more than one-third of the population experiencing food insecurity, the government has not effectively implemented either policy due to institutional disagreements and overlapping objectives (Shahristan, 2014). Street food vendors contribute substantially to the urban food supply in Afghanistan; however, they operate largely outside formal regulatory frameworks. These vendors often function under unhygienic conditions, and there is insufficient legal authority to regulate or monitor their activities. This reflects a broader gap in the adoption of internationally recognized food safety approaches, including risk-based inspections and hazard analysis systems (Khalid, 2016).

Several critical challenges remain within Afghanistan's food safety system. These include the need for harmonization of national laws with regional and international standards, improved coordination among regulatory authorities, and strengthening the independence

and effectiveness of inspection systems. In addition, there is a need for comprehensive training of inspectors, particularly in risk-based inspection methods, reform of food sampling strategies (especially for nonanimal origin and imported foods), enhanced surveillance of FBDs, stricter control of food adulteration, and support for the implementation of self-regulatory systems based on HACCP principles and good hygiene practices (Khalid, 2016).

There is a lack of a coordinated and functional system for monitoring FBDs, implying that cases such as hospitalizations due to food poisoning are not systematically recorded in a national database (Zahir & Hamdard, 2025). To address these issues, adopting a comprehensive food safety framework based on the One Health approach is recommended, along with fostering multisectoral collaboration to enhance awareness and engagement among stakeholders. Antibiotics are widely available without prescription, and their use in livestock production to promote growth contributes to the emergence of antimicrobial-resistant bacteria, which can be transmitted to humans through food. Notably, recent studies have reported 100% resistance to ampicillin among *Salmonella* isolates from chicken meat in Kabul (Ahmadi et al., 2024). Improving food safety and food security in Afghanistan, therefore, requires a multifaceted approach that addresses underlying political and socioeconomic constraints. This includes increased investment in agriculture, rural development, infrastructure, and initiatives that promote stability and long-term development (Zahir, Irfan, & Anis, 2025).

Economic constraints. Food insecurity remains one of the most pressing economic challenges for nations. Afghanistan's economic growth is sturdily influenced by the previous prolonged conflict, immigrants returning from neighboring countries, internal displacement, and climate change. Under such conditions, households are often forced to prioritize calorie intake over food safety. The situation has further deteriorated due to a reduction in humanitarian assistance. In Afghanistan, in 2025, 298 nutrition facilities that previously provided food aid were closed, and more than 141,000 individuals in southern Afghanistan lost access to assistance following the shutdown of 38 World Food Programme (WFP)-supported nutrition clinics (European Commission/WFP, 2025). Afghanistan's agricultural sector, which employs over 60% of the workforce, has been severely affected by drought, climate change, and ongoing economic instability. Postharvest losses and product spoilage frequently push farmers into financial hardship, resulting in land abandonment (Zahir et al., 2024).

In the livestock sector, 73% of herders have reduced their herd sizes due to the high costs of feed and veterinary services. Furthermore, Afghanistan's reliance on food imports, particularly wheat, makes it highly vulnerable to external price fluctuations. Increases in wheat prices in neighboring Pakistan significantly reduce affordability for Afghan consumers. This dependency also poses food safety concerns, as border closures or price shocks may force populations to rely on locally produced, nonstandard food products (Zahir & Hamdard, 2025). The economic crisis has intensified food safety risks. According to the WFP, funding shortages have driven many households to adopt "extreme coping" strategies. These include purchasing the cheapest available meat regardless of its safety or origin, and selling or consuming unrefrigerated milk to avoid starvation (European Commission/WFP, 2025). Overall, food insecurity and safety in Afghanistan, driven by conflict, economic instability, climate change, and reduced humanitarian aid, force households to prioritize survival over food safety. This has weakened the agricultural and livestock sectors, increased reliance on imports, and led to risky coping strategies, such as consuming low-quality or unsafe food, thereby intensifying food safety and public health challenges.

Control measures and policies

Existing efforts. Afghanistan's primary food safety regulator remains the Ministry of Public Health's National Food and Drug Authority (NFDA). In February 2026, the MoPH conducted three training sessions for its information and public relations staff on risk communication (RCCE), social and behavior change (SBC), and WASH. The Ministry continues to prioritize food safety education. A notable example is a one-day intensive training held in June 2025 for 50 teachers at Adei High School in Kabul. Targeting teachers is a strategic approach, as they act as knowledge multipliers by transferring food safety awareness to students, who in turn share it with their families and communities. Similarly, 300 bolani producers in Kabul participated in a three-day food safety training program (MoPH, 2025a; MoPH, 2025b).

The First Foods Afghanistan initiative, a joint flagship program by UNICEF, FAO, and WFP, launched in 2025 with support from the British government, represents a significant advancement in Afghanistan's food safety and nutrition landscape. This initiative exemplifies a system-based approach to addressing child malnutrition and food insecurity (UNICEF, 2026a). In parallel, UNICEF Afghanistan has implemented a vocational training program for approximately 1,000 women and youth across seven districts: Qaysar (Faryab), Pasaband (Ghor), Mohammad Agha (Logar), Lalpur (Nangarhar), Jani Khel (Paktika), and Qalat (Zabul). The program focuses on safe food production, preparation, and cooking practices, food storage, basic hygiene principles, and complementary feeding guidelines for children aged 6–23 months in alignment with WHO recommendations, along with business management skills to support economic empowerment (UNICEF, 2026b).

Despite an overall decline in humanitarian funding, targeted investments in nutrition and food safety continue. In February 2026, the MoPH signed a $492,328 agreement with the Afghan Youth Services Organization (AYSO) to enhance nutrition-related health services in Ghor and Badghis provinces through Basic Health Centers (MoPH, 2026). Additionally, a $100 million project implemented by FAO in collaboration with the Asian Development Bank aims to improve food and nutrition security and revitalize agricultural livelihoods. This initiative is expected to support approximately 151,000 households (around 1,057,000 individuals) over two years, with a focus on returnees, host communities, and populations affected by natural disasters such as earthquakes and floods, indicating that gaps in knowledge application persist (FAO, 2026).

Future perspectives

One Health approach. The One Health approach is founded on the principle that human, animal, and environmental health are interconnected and cannot be addressed in isolation. For a country like Afghanistan, this concept is particularly relevant, as foodborne pathogens circulate across livestock, humans, and the environment without regard for boundaries. A study highlighted this issue, recommending the formulation of a comprehensive food safety policy within a One Health framework (Zahir, Anis, & Mushfiq, 2025). At present, however, this integrated approach remains largely unimplemented. AMR surveillance is conducted separately in human and animal sectors, resulting in fragmented monitoring systems (Zahir & Hamdard, 2025). A fully operational One Health approach would establish a unified system capable of linking antibiotic use in livestock to treatment outcomes in human healthcare settings. Given that previous studies have already identified significant antibiotic resistance among foodborne pathogens (Ahmadi et al., 2024), such integration is not only beneficial but urgently required. In addition, researchers have emphasized the need for collaborative initiatives aimed at increasing awareness among all stakeholders involved in food safety. One promising step in this direction is the First Foods Afghanistan initiative, a joint program implemented by UNICEF, FAO, and WFP, representing a model of integrated action (UNICEF, 2026a).

The ongoing WASH crisis continues to undermine food safety efforts. Approximately 25% of households rely on unsafe water sources, while 37% lack access to basic soap for proper hand hygiene, creating conditions that facilitate contamination (World Vision, 2025). Recognizing these challenges, the First Foods initiative also addresses broader systemic risks, including the impacts of climate change on food systems, by promoting climate-resilient agriculture and sustainable dietary practices. These efforts aim to reduce long-term dependence on ultra-processed foods and imported staples. Ultimately, effective food safety systems must be built on resilient and integrated foundations rather than relying solely on regulatory frameworks that exist only on paper (UNICEF, 2026a).

Education programs. Future training programs must move beyond general awareness messages and instead emphasize practical, hands-on behavioral change. Studies identified a clear association between education level, possession of a health certificate, and actual food safety knowledge. These findings indicate that training programs are necessary on evidence-based and tailored accordingly. Specifically, programs should prioritize practical and visual training methods for food handlers with low literacy levels; rather than relying on lengthy written manuals, training should incorporate images, demonstrations, and hands-on practice. In addition, health certification systems must be made mandatory but also accessible, ensuring that individuals can obtain certification without high cost or administrative barriers. Regular refresher courses implementation is essential, as one-time training is insufficient to sustain safe practices. Furthermore, food safety education is to be integrated into existing vocational training programs (Zahir, 2026).

Research needs. Future research in Afghanistan should address several interconnected priorities to strengthen food safety systems and improve understanding of the burden and transmission of FBDs. A major research need exists in relation to surveillance gaps, as the current evidence base remains fragmented, underreported, and heavily dependent on cross-sectional or hospital-based studies with limited national representativeness. Strengthening surveillance systems requires the development of integrated national databases, standardized reporting frameworks, improved outbreak investigation systems, and expanded laboratory and diagnostic capacity, particularly in rural and conflict-affected areas where monitoring remains weak. There is also a critical need for longitudinal and nationwide epidemiological studies capable of generating reliable estimates of disease incidence, mortality, DALYs, economic burden, and population-level exposure risks associated with FBDs in Afghanistan.

In addition, research should focus on pathogen-specific investigations to better characterize the prevalence, transmission dynamics, molecular epidemiology, seasonal variation, and environmental persistence of major bacterial, viral, parasitic, and zoonotic pathogens, including *Salmonella* spp., *Shigella* spp., *Campylobacter* spp., *E. coli*, *V. cholerae*, *G. intestinalis*, *T. gondii*, *L. monocytogenes*, and other emerging food- and waterborne agents circulating through informal food systems, livestock production chains, and contaminated water sources. Further molecular and genomic studies are needed to identify transmission pathways, outbreak sources, and pathogen evolution within Afghanistan's complex food and environmental systems.

Antimicrobial resistance (AMR) monitoring also represents a major research priority due to the widespread misuse of antibiotics in both human and livestock sectors and the limited implementation of antimicrobial stewardship programs. Future studies should establish coordinated One Health AMR surveillance systems linking human health, veterinary medicine, food production, and environmental sectors to monitor antibiotic usage patterns, resistant foodborne pathogens, and transmission routes across the food chain. Research is also required to evaluate the effectiveness of current NAPs, veterinary oversight systems, and interventions aimed at reducing inappropriate antimicrobial use in livestock production. Beyond microbial hazards, important knowledge gaps remain regarding chemical hazards in the Afghan food supply. Future investigations should examine the occurrence and health impacts of food adulteration, pesticide residues, heavy metals, naturally occurring toxins such as pyrrolizidine alkaloids, veterinary drug residues, and other environmental contaminants in food and water. Comprehensive food toxicology studies and chemical risk assessments are needed to better understand exposure pathways and long-term health consequences associated with contaminated food products.

Water, sanitation, and hygiene (WASH)-related risks also require further investigation, particularly because contaminated water, poor sanitation infrastructure, open defecation, and inadequate hygiene practices continue to sustain the transmission of enteric and parasitic infections. Future research should evaluate regional differences in WASH conditions, identify vulnerable populations at highest risk, assess the microbiological quality of drinking and irrigation water, and determine the effectiveness of WASH interventions in reducing diarrheal and parasitic disease burdens. Greater attention should also be given to understanding the combined foodborne and waterborne transmission pathways that contribute to persistent fecal–oral disease transmission in Afghanistan. In addition, more comprehensive evidence is needed regarding food system interventions and structural food safety improvements.

Future studies should assess the effectiveness of hygiene education programs, behavioral change interventions, food handler training, risk-based inspections, HACCP implementation, cold chain expansion, improved transportation and storage systems, and regulation of informal food markets. Research should further investigate the impact of socioeconomic instability, climate change, population displacement, food insecurity, and infrastructure limitations on food safety outcomes and contamination risks across the food supply chain. Finally, future work should prioritize integrated One Health-based approaches that strengthen coordination between public health, agriculture, veterinary, environmental, and food regulatory sectors, while also evaluating policy reforms, institutional coordination mechanisms, and multisectoral food safety governance strategies capable of improving resilience, reducing contamination risks, and supporting evidence-based policymaking in Afghanistan.

Strengths and limitations of the review. A major strength of this review is its comprehensive synthesis of evidence from Afghanistan and comparable LMICs, offering a broader understanding of food safety challenges in fragile and resource-constrained settings. However, this review is also subject to several limitations inherent in the available literature. Much of the evidence is derived from cross-sectional studies conducted in geographically restricted areas, frequently relying on convenience sampling and relatively small sample sizes. Furthermore, fragmented surveillance systems, limited geographical coverage, and inconsistencies in diagnostic methods and reporting practices may have affected the accuracy and comparability of prevalence estimates. These methodological limitations constrain the ability to establish causal relationships or generate nationally representative estimates. Despite these challenges, the consistency of findings across diverse studies provides valuable insights into the key food safety issues affecting Afghanistan and other resource-limited settings, highlighting important priorities for future research, surveillance, and policy interventions.

Conclusion

In conclusion, FBDs remain a major yet underrecognized public health challenge in Afghanistan, driven by weak surveillance systems, fragmented regulation, poor sanitation, informal food markets, antimicrobial misuse, and limited food safety infrastructure. Persistent gaps in laboratory and diagnostic capacity, underreporting of outbreaks, and insufficient nationwide epidemiological data continue to hinder accurate assessment of disease burden and delay effective response measures. Although these challenges are common across LMICs,

Afghanistan faces a more difficult situation due to prolonged instability, fragile institutions, weak public health systems, and limited food safety infrastructure.

The evidence also highlights a persistent knowledge–practice gap among food handlers and consumers, where basic food safety awareness is not consistently translated into safe practices because of limited practical training, economic constraints, and weak regulatory oversight. Infrastructure limitations, including inadequate cold chain systems, poor transportation conditions, unreliable electricity, and limited laboratory capacity, further intensify contamination risks and delay outbreak detection. In addition, antimicrobial resistance is emerging as a serious food safety threat, closely linked to uncontrolled antibiotic use, weak veterinary oversight, and insufficient antimicrobial stewardship in food-animal production systems. Poor WASH conditions and limited regulation of informal food markets further sustain pathogen transmission and contamination risks.

Addressing these challenges requires realistic, context-specific, and evidence-based food system interventions. Priority actions should include strengthening integrated surveillance systems, expanding pathogen-specific and AMR monitoring, improving laboratory and diagnostic capacity, implementing risk-based inspections, enhancing hygiene education and food handler training, improving WASH infrastructure, expanding cold storage and transport systems, and strengthening coordination between human, animal, environmental, and food regulatory sectors through a One Health approach. Future research should prioritize surveillance gaps, pathogen-specific investigations, chemical hazard assessments, WASH-related risks, and evaluation of food system interventions to support resilient and sustainable food safety policymaking in Afghanistan and other fragile LMIC settings. Overall, improving food safety requires not only policy development but also sustained investment in infrastructure, institutional coordination, and effective implementation of practical, risk-based, and One Health-oriented strategies to reduce the burden of FBDs and strengthen public health resilience.

Ethical approval

In this study, no animals or humans were involved.

Consent to publication

All authors have given their full consent for the submission of this research article in this journal.

Declaration of generative AI in scientific writing

During the preparation of this work, the authors used ChatGPT to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the published article.

Availability of data and material

Not applicable.

CRediT authorship contribution statement

Ahmadullah Zahir: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Conceptualization. **Mohammad Aman Ahmadzai:** Writing – original draft, Methodology, Funding acquisition, Formal analysis, Data curation. **Mohammad Naeem Azizi:** Writing – original draft, Visualization, Validation, Software, Resources, Project administration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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